

Splunk SPL Query Collection

SOC Alert Triage & Investigation Dashboard Project

Overview

This document provides an overview of the Splunk Processing Language (SPL) queries developed during the SOC Alert Triage & Investigation Dashboard project. The queries were created in Splunk Enterprise to support authentication monitoring, threat hunting, suspicious web request analysis, persistence detection, automated scanner detection, and security event correlation within a simulated SOC investigation environment.

Investigation Areas

- SSH brute-force detection
- Failed authentication monitoring
- Successful login correlation
- Suspicious web request analysis
- HTTP status code analysis
- Persistence detection
- Automated scanner and reconnaissance detection
- Threat severity classification
- Security event correlation

Skills Demonstrated

- SPL query development
- SIEM dashboard creation
- Authentication monitoring
- Threat hunting
- Incident triage
- Log correlation
- Security monitoring
- Web request analysis
- Persistence investigation
- Threat severity prioritization

Project Purpose

The purpose of this project was to simulate SOC investigation workflows using Splunk Enterprise SIEM. The dashboard and SPL queries were designed to improve visibility into suspicious authentication behavior, web application attacks, persistence-related activity, and automated reconnaissance attempts through centralized log analysis and visualization techniques.

Conclusion

The completed SPL query collection demonstrates practical experience using Splunk Enterprise for SOC alert triage and cybersecurity investigations. The project combines log analysis, threat detection, event correlation, and dashboard visualization techniques commonly used in Security Operations Center environments.